

Feynman Path Integrals Example

Compute the kernel for a free particle

$$L = \frac{m}{2} \dot{x}^2 \quad : \text{Lagrangian of a free particle!}$$

Using

$$A = \left(\frac{2\pi i \hbar \varepsilon}{m} \right)^{1/2}$$

$$K(b, a) = \lim_{\varepsilon \rightarrow 0} \frac{1}{A} \int \dots \int e^{(i/\hbar) S[b, a]} \frac{dx_1}{A} \frac{dx_2}{A} \dots \frac{dx_{N-1}}{A}$$

with

$$S[b, a] = \int_{t_a}^{t_b} L(\dot{x}, x, t) dt.$$

we have

$$K(b, a) = \lim_{\varepsilon \rightarrow 0} \left(\frac{2\pi i \hbar \varepsilon}{m} \right)^{-N/2} \int \dots \int \exp \left\{ \frac{im}{2\hbar\varepsilon} \sum_{j=1}^N (x_j - x_{j-1})^2 \right\} dX, \quad dX = dx_1 dx_2 \dots dx_{N-1}.$$

$\Rightarrow$ this is the set of Gaussian integrals $(\int e^{-\alpha x^2} dx)$.

Sample

$$\left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \int_{-\infty}^{\infty} \exp \left\{ \frac{im}{2\hbar\varepsilon} [(x_2 - x_1)^2 + (x_1 - x_0)^2] \right\} dx_1$$

Expand

$$\begin{aligned} (x_2 - x_1)^2 + (x_1 - x_0)^2 &= x_2^2 - 2x_2x_1 + x_1^2 + x_1^2 - 2x_1x_0 + x_0^2 \\ &= (x_2^2 + x_0^2) + 2x_1^2 - 2(x_2 + x_0)x_1. \end{aligned}$$

$$\left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \exp \left\{ \frac{im}{2\hbar\varepsilon} \frac{(x_2 - x_0)^2}{2} \right\}$$

Let

$$\Phi(x_1) = \frac{im}{2\hbar\varepsilon} [(x_2^2 + x_0^2) + 2x_1^2 - 2(x_2 + x_0)x_1].$$

$$\int_{-\infty}^{\infty} e^{-ax^2+bx+c} dx = \sqrt{\frac{\pi}{a}} \exp \left(\frac{b^2}{4a} + c \right),$$

$$a = -\frac{im}{2\hbar\varepsilon} \cdot 2 = -\frac{im}{\hbar\varepsilon}, \quad b = -\frac{im}{2\hbar\varepsilon} (-2(x_2 + x_0)) = \frac{im}{\hbar\varepsilon} (x_2 + x_0),$$

$$c = \frac{im}{2\hbar\varepsilon}(x_2^2 + x_0^2).$$

$$\int_{-\infty}^{\infty} e^{\Phi(x_1)} dx_1 = \sqrt{\frac{\pi\hbar\varepsilon}{im}} \exp\left\{\frac{im}{4\hbar\varepsilon}(x_2 - x_0)^2\right\}.$$

Iterating the Gaussian integrals over all slices gives

$$K(x_b, t_b; x_a, t_a) = \left(\frac{m}{2\pi i\hbar T}\right)^{1/2} \exp\left\{\frac{im}{2\hbar T}(x_b - x_a)^2\right\}, \quad T = t_b - t_a.$$

From the single-slice integral we had

$$I = \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{1/2} \int_{-\infty}^{\infty} \exp\left\{\frac{im}{2\hbar\varepsilon}[(x_2 - x_1)^2 + (x_1 - x_0)^2]\right\} dx_1.$$

Completing the square and using the Gaussian formula gives

$$\begin{aligned} I &= \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{1/2} \left[\sqrt{\frac{\pi\hbar\varepsilon}{im}} \exp\left\{\frac{im}{2\hbar(2\varepsilon)}(x_2 - x_0)^2\right\} \right] \\ \Rightarrow I &= \left(\frac{m}{2\pi i\hbar(2\varepsilon)}\right)^{1/2} \exp\left\{\frac{im}{2\hbar(2\varepsilon)}(x_2 - x_0)^2\right\}. \end{aligned}$$

Next multiply result I by

$$\left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{1/2} \exp\left\{\frac{im}{2\hbar\varepsilon}(x_3 - x_2)^2\right\}$$

and continue to integrate, giving

$$\left(\frac{m}{2\pi i\hbar(3\varepsilon)}\right)^{1/2} \exp\left\{\frac{im}{2\hbar(3\varepsilon)}(x_3 - x_0)^2\right\}.$$

Follow by induction for n steps (recursion) to get

$$\left(\frac{m}{2\pi i\hbar(n\varepsilon)}\right)^{1/2} \exp\left\{\frac{im}{2\hbar(n\varepsilon)}(x_n - x_0)^2\right\}, \quad n\varepsilon \text{ is the discrete time.}$$

So after $N - 1$ steps we get

$$K(b, a) = \left(\frac{m}{2\pi i\hbar(t_b - t_a)}\right)^{1/2} \exp\left\{\frac{im}{2\hbar(t_b - t_a)}(x_b - x_a)^2\right\}. \quad \square$$